

Churn Prediction & Retention Strategy

A Data-Driven Business Proposal for Company A

Telecommunications Industry

Executive Summary

01

The Problem

Company A shows a 49.6% churn rate within a 31–60 day observation window — well above the industry average of 15–25%. Even accounting for the short observation window, the data signals a systemic retention challenge with \$34.4M in annual revenue at risk.

03

The Model

Two ML classifiers compared: Logistic Regression (AUC 0.6151) and Random Forest (AUC 0.6677). Random Forest selected as an early-warning ranking tool. Deployment uses a 0.5 probability threshold to prioritize at-risk customers.

02

The Data

100,000 customer records across 100 features. Two datasets merged: Client (demographics, device, account) and Record (usage metrics, revenue). Target variable: churn (0 = retained, 1 = churned within observation window).

04

The Proposal

Targeted retention program for 3,976 high-risk customers (old devices + declining usage, flagged by model at threshold=0.5). Conservative estimates project \$337K net annual benefit at 424% ROI, pending pilot validation.

Market Context: The Telecom Churn Crisis

15–25%

Industry-average
annual churn rate

McKinsey, 2023

\$100B+

Annual revenue lost
globally to churn

Bain & Co., 2022

5–7×

Cost to acquire vs
retain a customer

Harvard Business Review

5% ↓

Churn reduction
raises profit 25–95%

Bain & Co.

Why It Matters for Company A

Company A's dataset shows a 49.6% churn indicator within a 31–60 day observation window — not a calendar-year attrition rate, but a near-term behavioral signal that far exceeds industry benchmarks. In telecom, customers increasingly churn due to device experience and service quality rather than pricing alone, making behavioral prediction a high-value intervention tool.

Dataset Overview: Company A Telecom Data

Client.csv — Customer Profile

100,000 rows × 50 columns

Demographic and account data: device age (eqpdays), device price (hnd_price), household attributes (income, adults, dwelling type), credit class, tenure in months, subscriber counts.

Record.csv — Usage History

100,000 rows × 51 columns

Behavioral metrics: monthly revenue (rev_Mean), usage minutes (mou_Mean), call completion rates, overage charges, roaming, customer care calls, and usage trend (change_mou).
Target variable: churn (0 = retained · 1 = churned within 31–60 day window)

Merged on Customer_ID → 100,000 rows × 100 columns | Imputed with median strategy | 80/20 stratified train-test split

EDA: Churn Rate & Revenue at Risk

49.6%	\$57.91	\$694.97	\$34.4M
Churn Rate	Avg Monthly Revenue	Annual LTV (est.)	Annual Revenue at Risk
49,562 of 100,000 customers	per customer	per customer	from churning cohort

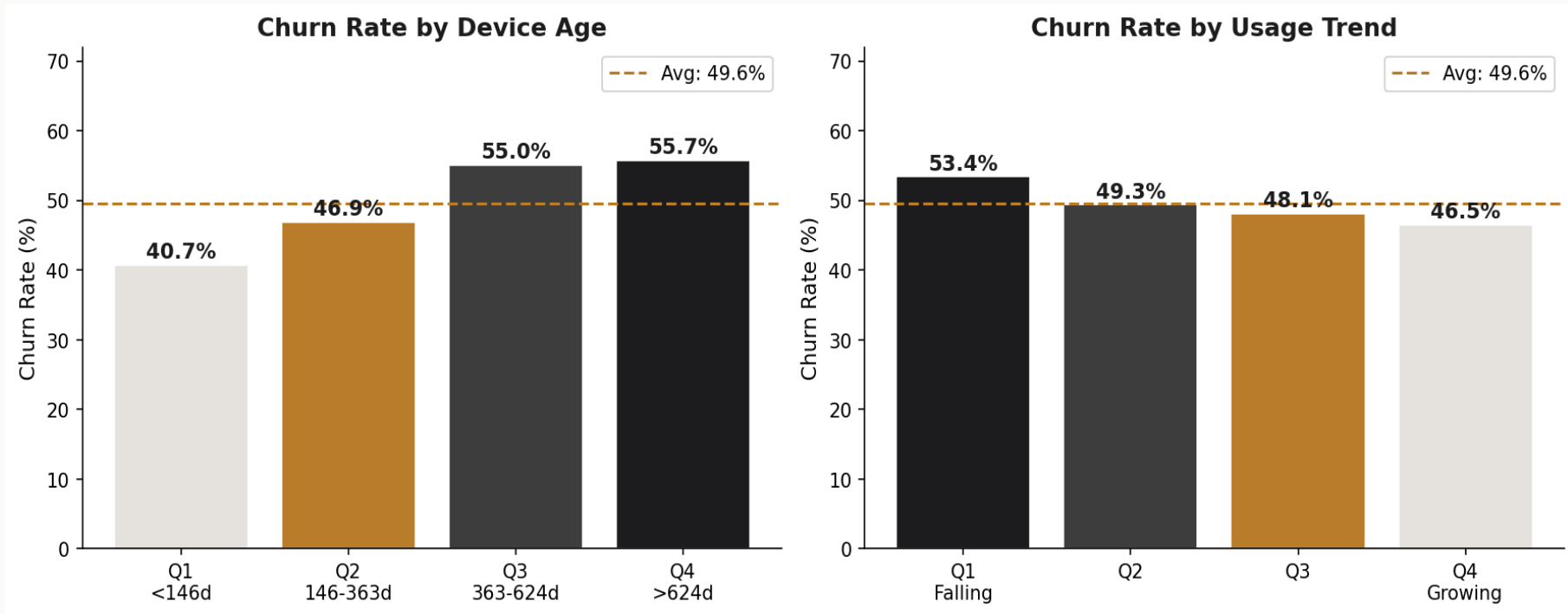


Note on Churn Definition

The churn variable captures whether a customer churned within a 31–60 day window after the observation date — not over a full calendar year. A 49.6% rate therefore does not imply the company loses half its customers annually, but that roughly half the observed cohort showed a short-term churn event.

This framing is intentional: the business value of the model lies in identifying near-term churn signals early enough to allow proactive intervention. Revenue at risk figures are calculated on an annualized basis for strategic comparability.

EDA: Key Churn Drivers



Device Age: 55.7% churn (oldest) vs 40.7% (newest) · #1 predictor **Usage Decline:** 53.4% churn (falling) vs 46.5% (growing) · Observable before churn occurs

Problem Definition & ML Approach

Business Problem

Core question: Which customers are most likely to churn in the next 31–60 days?

Objective: Build a model that identifies at-risk customers early enough to enable proactive retention.

Target variable: churn (binary: 0 = retained, 1 = churned)

Task type: Binary classification

Justification: EDA confirms churn follows behavioral signals (device age, usage decline) observable before the event. A classifier can rank customers by churn probability for targeted outreach.

ML Approach

1 Data Preparation

Merge datasets. Median imputation. 80/20 stratified split.

2 Feature Engineering

4 engineered features: completion_rate, rev_per_mou, overage_ratio, high_risk flag.

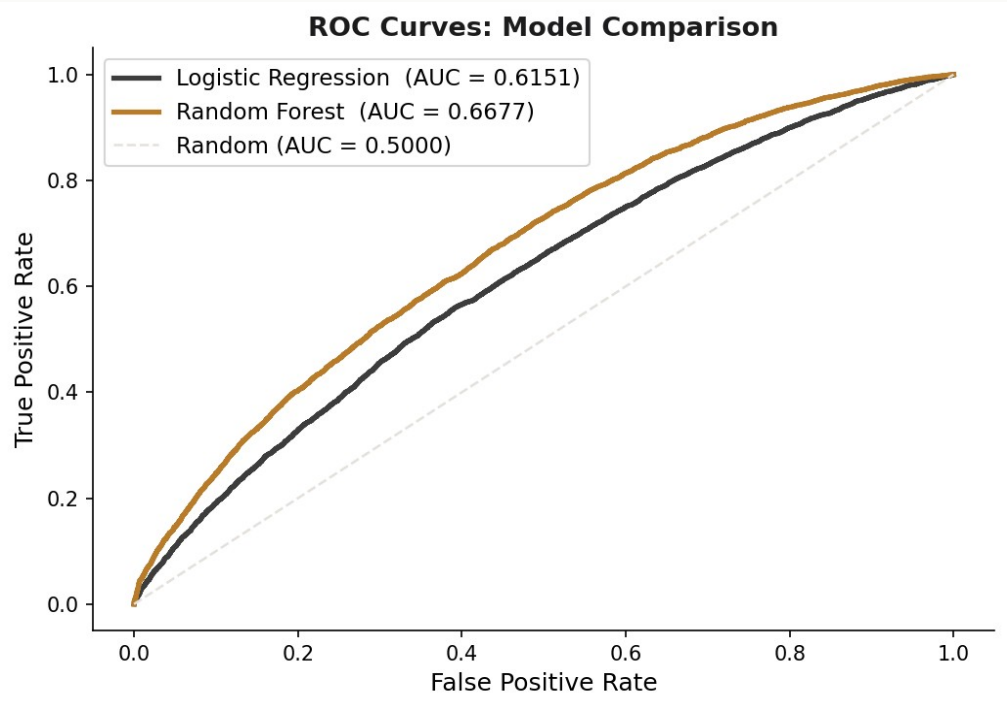
3 Model Training

LR (StandardScaler + balanced weights) vs RF (100 trees, max_depth=10, balanced).

4 Evaluation

AUC-ROC for ranking quality. Threshold=0.5 for deployment-ready flagging.

Model Results: Performance Comparison



Performance Summary

Model	AUC-ROC	Accuracy
Logistic Regression	0.6151	58.2%
Random Forest ✓	0.6677	61.5%

Selected: Random Forest

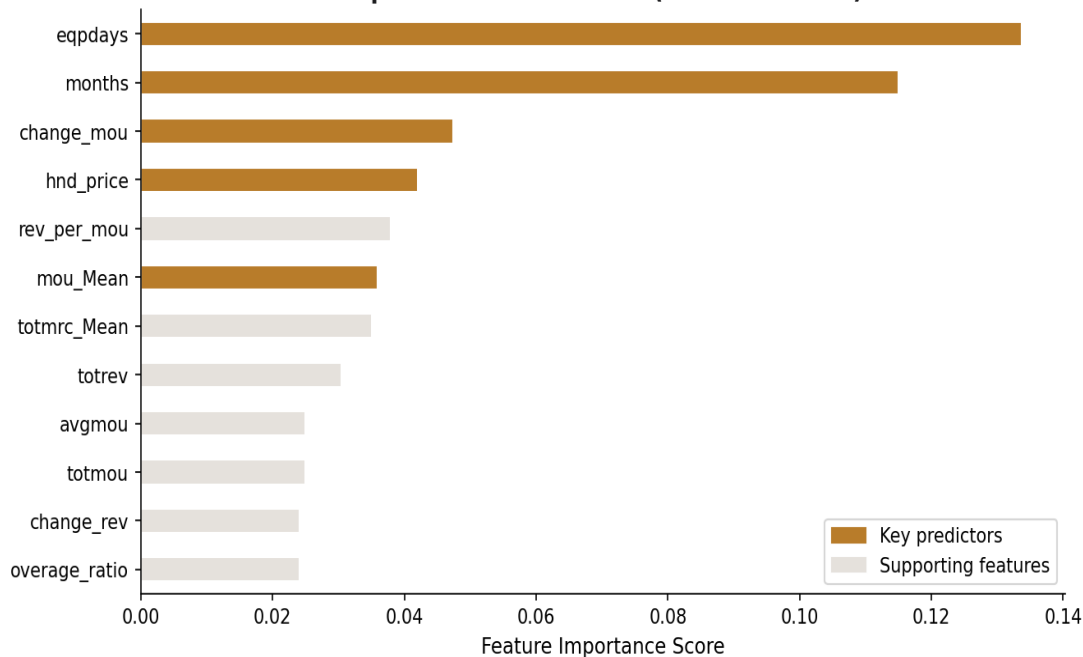
Higher AUC-ROC — better at ranking customers by churn probability

- Captures non-linear feature interactions
- Feature importance directly interpretable for business stakeholders
- At threshold=0.5: flags 55,472 customers across full base

Note: AUC-ROC of 0.6677 positions the model as a useful early-warning ranking tool — not a perfect predictor. The value lies in prioritizing outreach, not eliminating uncertainty.

Model Insights: What Drives Churn?

Top 12 Churn Predictors (Random Forest)



Business Interpretation

eqpdays 13.7%

Device age is the strongest predictor. Customers with older handsets churn significantly more — likely seeking upgrades from competitors.

months 11.6%

Tenure is non-linear. Mid-tenure customers (1–3 years) show the highest churn as initial contracts expire.

change_mou 5.1%

Declining usage is a leading indicator — disengagement is detectable weeks before the customer leaves.

hnd_price 4.0%

Customers with higher-priced devices have elevated expectations. Unmet service quality accelerates churn in this segment.

Business Proposal

Proactive Retention Program for High-Risk Customers

01

Identify

Apply Random Forest at threshold=0.5 monthly. Focus intervention on the high-risk behavioral segment: eqpdays > Q3 AND change_mou < Q1. Current target: 3,976 customers flagged.

02

Intervene

Deploy targeted retention: device upgrade offers, loyalty credits, and proactive customer service calls — timed before the customer initiates churn, not after.

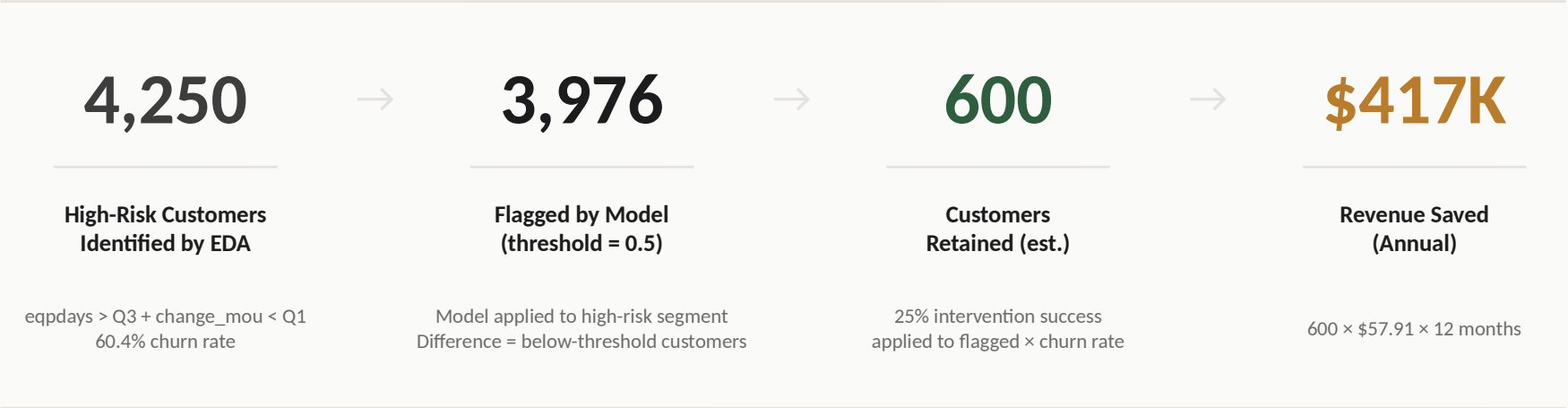
03

Measure

Track churn rate in the flagged cohort vs a hold-out control group. Monitor retained revenue and ROI monthly. Retrain model quarterly with new behavioral data.

Quantified Business Impact

Assumptions: High-risk = eqpdays > Q3 AND change_mou < Q1 · Model threshold=0.5 · 60.4% segment churn rate · 25% intervention save rate · \$57.91 avg monthly revenue · \$20/customer/year program cost



Annual Revenue Saved	Program Cost	Net Annual Benefit	ROI
\$416,980	\$79,520	\$337,460	424%

Implementation Roadmap

Phase 1 · Month 1

Model Deployment

Integrate RF model into CRM pipeline

Score all customers monthly at threshold=0.5

Flag high-risk behavioral segment

Define intervention playbook

Phase 2 · Month 2-3

Pilot Campaign

Target 500 highest-risk customers

Device upgrade discount or loyalty credit

Proactive calls for top 100 customers

Track conversion vs control group

Phase 3 · Month 4-6

Scale & Optimize

Roll out to full 3,976 customer segment

A/B test intervention types

Retrain model quarterly

Report ROI to executive stakeholders

Limitations & Analytical Boundaries

Acknowledging constraints is part of rigorous business analysis.

01 Observation Window

Churn captures events within 31–60 days, not calendar-year attrition. The 49.6% rate reflects near-term behavioral signals — extrapolating to annual loss would be misleading.

02 Moderate Predictive Power

AUC-ROC of 0.6677 is meaningful but limited. The model ranks customers by churn probability — it is a prioritization tool, not a definitive predictor of individual events.

03 ROI Assumptions Not Validated

The 424% ROI relies on a 25% intervention save rate (Bain & Co. reference) and \$20/customer cost. Both are conservative estimates pending empirical validation via the Phase 2 pilot.

04 Correlation, Not Causation

The model identifies associations. Device age predicts churn but may not cause it. Root causes require qualitative investigation before designing the intervention offer.

05 No Historical Retention Data

No past campaign data was available to calibrate save rates. The A/B test pilot (Phase 2 of roadmap) is required to validate effectiveness before full-scale deployment.

Conclusion

01 A measurable retention problem

Company A's data reveals a 49.6% near-term churn rate — significantly above industry benchmarks. \$34.4M in annual revenue is at risk from the current churning cohort.

02 Device age and usage decline are the actionable signals

eqpdays and change_mou are the strongest predictors — both observable before churn occurs. They give the business a lead time window for intervention.

03 The model as a ranking and prioritization tool

Random Forest (AUC 0.6677, Accuracy 61.5%) enables reliable ranking of at-risk customers. At threshold=0.5, it identifies 3,976 highest-priority customers for immediate action.

04 Retention could deliver measurable ROI, pending pilot validation

Conservative scenario: \$337K net annual benefit, 424% ROI. Projections are scenario-based and require validation via a controlled pilot before full-scale deployment.

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